



High-Resolution Imagery for Disaster Mapping

Design and Development of UAV-Based Monitoring Frameworks

Module 04 - Topic #3

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of West Attica)

DIGIHAZ is implemented by:



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Learning Objectives

By the end of this lecture you will be able to:

- 01 Formulate Geohazard-Specific Strategies
- 02 Learn about Open Data standards
- 03 Design Emergency Integration Workflows

Topics Covered

- Geohazard-Specific UAV Monitoring Solutions
- Open Software Data Processing and Interoperability Standards
- Integration with Civil Protection Emergency Infrastructures

Estimated Duration
45 minutes lecture + 45 min exercise

Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Need to tailor UAV solutions to each geohazard
- Interoperable data processing pipelines
- Integration to civil protection infrastructure

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Geohazard-Specific UAV Monitoring Solutions

- Landslides and Mass movements

Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Landslides and Mass movements
- Accuracy of up to 0.1m for 3D displacement fields

Drone Imagery Processing and Photogrammetric Workflows

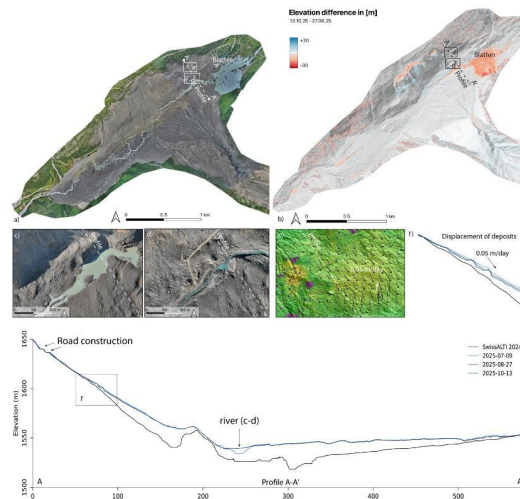
Geohazard-Specific UAV Monitoring Solutions

- Landslides and Mass movements
- Accuracy of up to 0.1m for 3D displacement fields
- Optimal framework for landslides:
A «hybrid approach»
 - Fixed-wing LiDAR
 - Multicopter UAV for fine-scale inspection

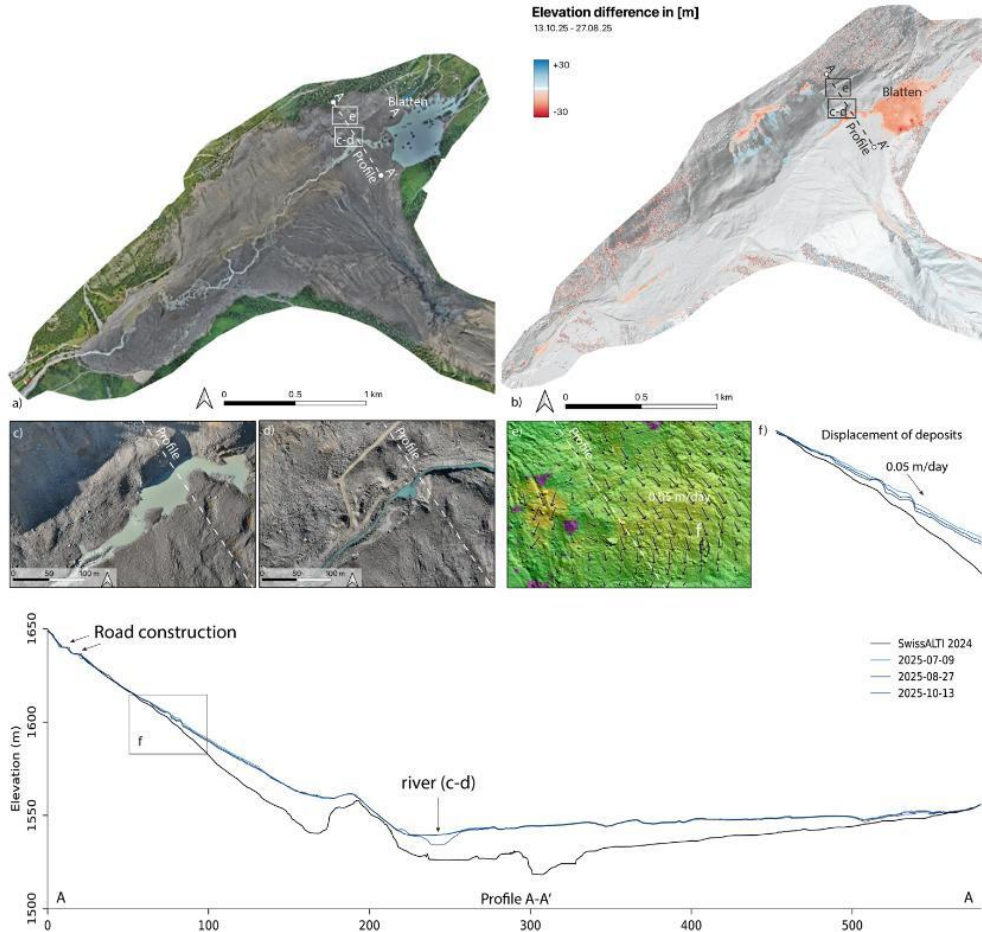
Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Landslides and Mass movements
- Accuracy of up to 0.1m for 3D displacement fields
- Optimal framework for landslides:
A «hybrid approach»
 - Fixed-wing LiDAR
 - Multirotor UAV for fine-scale inspection
- Ever-increasing use



Monitoring Workflows



Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Floods and Inundations

Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Floods and Inundations
- DTMs and hydrological simulations

Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Floods and Inundations
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- «Extrapolated Flood Surface» framework (EXFLOS)

Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Floods and Inundations
- DTMs and hydrological simulations
- «Extrapolated Flood Surface» framework (EXFLOS)
- Provides reliable flood extent estimates

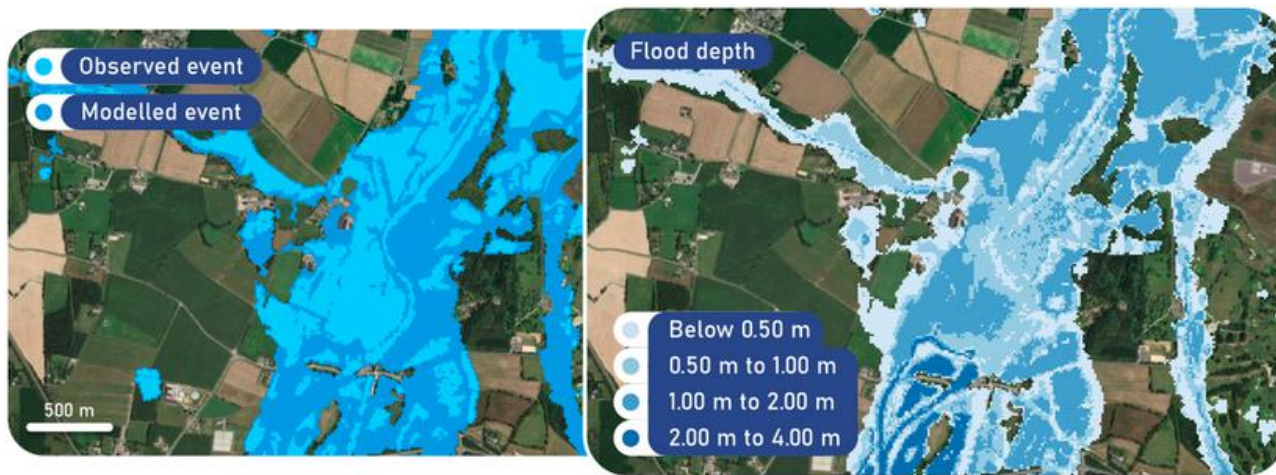
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Geohazard-Specific UAV Monitoring Solutions

- Floods and Inundations
- DTMs and hydrological simulations
- «Extrapolated Flood Surface» framework (EXFLOS)
- Provides reliable flood extent estimates
- Uses gravitational parameters and topographic depressions

Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions



Example of flood depth product, generated with EXFLOS for January 2025 flood in Ille-et-Vilaine, France. Flood depth estimates, simulated based on the observed event, and expanded into concealed areas, reveal a more realistic flood extent.

Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Earthquakes and Volcanic Eruptions

Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Earthquakes and Volcanic Eruptions
- Priority to Structural integrity & thermal mapping

Drone Imagery Processing and Photogrammetric Workflows

Geohazard-Specific UAV Monitoring Solutions

- Earthquakes and Volcanic Eruptions
- Priority to Structural integrity & thermal mapping
- Orthomosaics & InSAR to map fault ruptures, surface displacements, structural collapse

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards

- «Successful response starts with a map»

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards

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- Rapid, functional dissemination is crucial

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards

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- OGC & interoperability protocols

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards

- «Successful response starts with a map»
- Rapid, functional dissemination is crucial
- OGC & interoperability protocols
- Guarantee for sharing data between «software ecosystems» without problems

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards

- Web Map Service (WMS)

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards

- Web Map Service (WMS)
- Web Feature Service (WFS)

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards

- Web Map Service (WMS)
- Web Feature Service (WFS)
- Web Coverage Service (WCS)

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards

- OGC Sensor Web Enablement (SWE) framework / SensorML

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Open Software Data Processing and Interoperability Standards

- OGC Sensor Web Enablement (SWE) framework / SensorML
- SensorML: drone capabilities in a standard XML format

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards

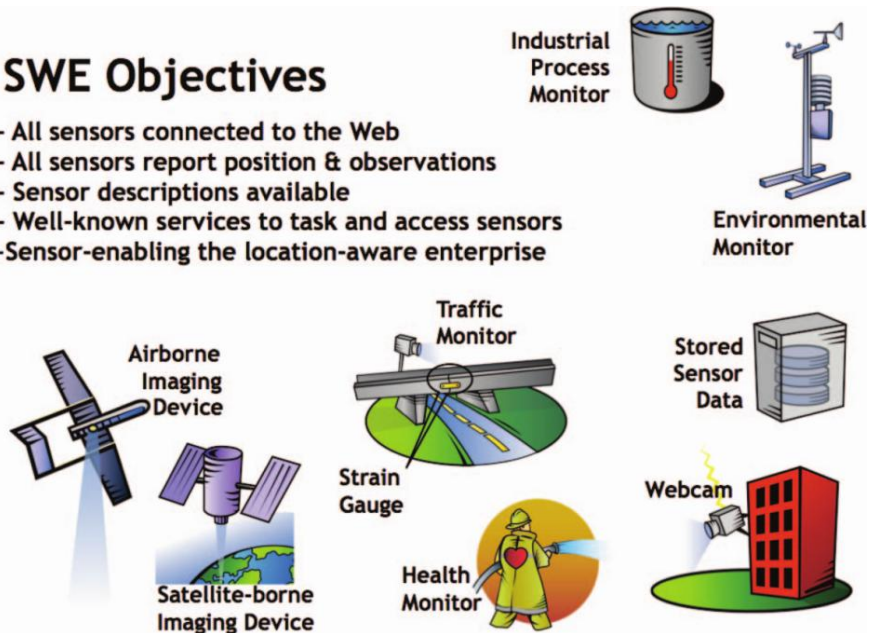
- OGC Sensor Web Enablement (SWE) framework / SensorML
- SensorML: drone capabilities in a standard XML format
- Geography Markup Language (GML)
- Keyhole Markup Language (KML)

Drone Imagery Processing and Photogrammetric Workflows

- OGC Sensor Web Enablement (SWE) framework / SensorML
- SensorML: drone capabilities in a standard XML format
- Geography Markup Language (GML)
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SWE Objectives

- All sensors connected to the Web
- All sensors report position & observations
- Sensor descriptions available
- Well-known services to task and access sensors
- Sensor-enabling the location-aware enterprise



Sensor Web Enablement (SWE) standards provide a platform for describing, publishing, discovering, controlling and collecting data from virtually any kind of sensor

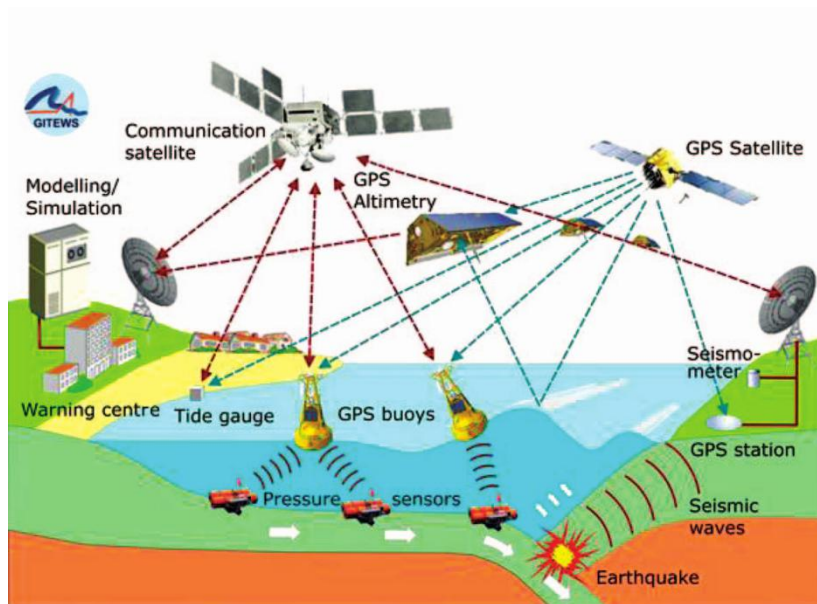
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Open Software Data Processing and Interoperability Standards

- Numerous real-world uses!
- Monitoring & control system in Wupper river, Germany
- Advance Fire information System in South Africa

Drone Imagery Processing and Photogrammetric Workflows

Open Software Data Processing and Interoperability Standards



- Numerous real-world uses!
- Monitoring & control system in Wupper river, Germany
- Advance Fire information System in South Africa
- Tsunami Early Warning system (GITEWS)

Drone Imagery Processing and Photogrammetric Workflows

Integration with Civil Protection Emergency Infrastructures

- UAV info & products must feed civil protection infrastructures

Drone Imagery Processing and Photogrammetric Workflows

Integration with Civil Protection Emergency Infrastructures

- UAV info & products must feed civil protection infrastructures
- ELDIS: A central information hub

Drone Imagery Processing and Photogrammetric Workflows

Integration with Civil Protection Emergency Infrastructures

- UAV info & products must feed civil protection infrastructures
- ELDIS: A central information hub
- Dispatcher-controller paradigm

Drone Imagery Processing and Photogrammetric Workflows

Integration with Civil Protection Emergency Infrastructures

- Common Alerting Protocol (CAP)

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Integration with Civil Protection Emergency Infrastructures

- Common Alerting Protocol (CAP)
- Ensures consistency across different medias

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Integration with Civil Protection Emergency Infrastructures

- Common Alerting Protocol (CAP)
- Ensures consistency across different medias
- Encoding of UAV-based estimates to CAP

Drone Imagery Processing and Photogrammetric Workflows

Integration with Civil Protection Emergency Infrastructures

- Global Disaster Alert and Coordination System (GDACS)



Hazard estimation & Impact Estimation under the GDACS framework.

Drone Imagery Processing and Photogrammetric Workflows

Integration with Civil Protection Emergency Infrastructures

- Global Disaster Alert and Coordination System (GDACS)
- Exchange & analysis of information between different international actors

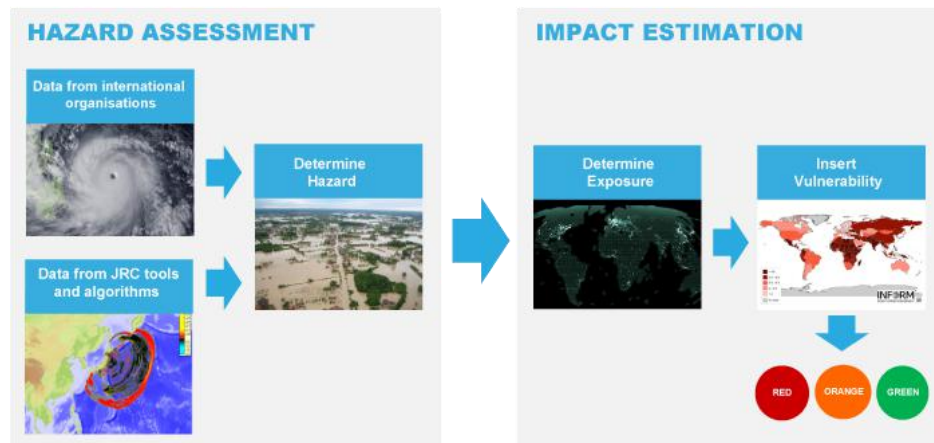


Hazard estimation & Impact Estimation under the GDACS framework.

Drone Imagery Processing and Photogrammetric Workflows

Integration with Civil Protection Emergency Infrastructures

- Global Disaster Alert and Coordination System (GDACS)
- Exchange & analysis of information between different international actors
- GDACS aims at reducing problems with possible duplication, gaps, overlap, etc.



Hazard estimation & Impact Estimation under the GDACS framework.

Drone Imagery Processing and Photogrammetric Workflows

Integration with Civil Protection Emergency Infrastructures

- Global Disaster Alert and Coordination System (GDACS)
- Exchange & analysis of information between different international actors
- GDACS aims at reducing problems with possible duplication, gaps, overlap, etc.
- Virtual On-site Operations Coordination Centre

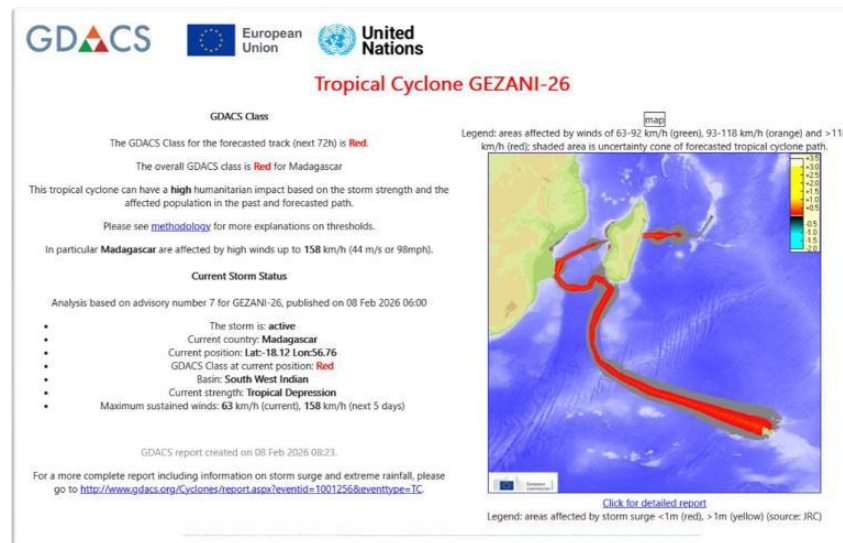


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Integration with Civil Protection Emergency Infrastructures

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Example of GDACS notification message by email.

Key terms & definitions

Key Terms & Definitions	
Sensor Model Language (SensorML)	An Open Geospatial Consortium (OGC) encoding standard that uses XML to describe the specific capabilities, calibration parameters, and outputs of sensors, enabling autonomous discovery and "plug-and-play" interoperability for drone payloads within larger emergency networks.
Web Map Service (WMS)	A standardized HTTP interface that allows for the requesting and sharing of geo-registered map images from distributed databases. In disaster management, it ensures that orthomosaics generated by drones in the field can be instantly visualized as layers on centralized command center dashboards.
Common Alerting Protocol (CAP)	An international, XML-based standard format for exchanging all-hazard emergency alerts. It allows emergency managers to take spatial hazard boundaries mapped by UAVs (like an expanding flood or chemical plume) and use them to trigger geographically targeted, consistent warnings simultaneously across mobile phones, internet feeds, and public broadcast systems.

Key terms & definitions

Key Terms & Definitions	
Global Disaster Alert and Coordination System (GDACS)	A cooperation framework between the United Nations and the European Commission that issues automated real-time disaster notifications, provides risk analysis, and operates a virtual coordination platform where international responders can exchange localized maps and UAV damage assessments.
Extrapolated Flood Surface (EXFLOS)	A specialized hydraulic modeling methodology that combines observed flooded areas with precise Digital Terrain Models (DTMs). By mathematically simulating how water distributes across topographic depressions, it allows analysts to accurately map flood extents even when the water is concealed beneath dense urban infrastructure or tree canopies.

Unmanned Systems & Image Processing Basics

Discussion questions

- 1.Name disaster types where you learned that UAV imagery is useful. In what ways does it provide use to experts and rescue teams?
- 2.Why are open standards important?
- 3.To what degree has the integration of UAV-based solutions with civil protection emergency infrastructures advanced?

Sneak peak at your exercise

Example

In the exercise for this topic, you will be tasked to write code that opens a GeoJSON file.

See also:

GitHub repo / Notebook link

1. Use Regular expressions, Python JSON libraries or other means to write a Python class to create basic functionality that manipulates a GeoJSON file.
2. The class should have at least methods to: read a GeoJSON from a file, write the GeoJSON to the file, create a new GeoJSON. Optionally include methods to manipulate the GeoJSON contents.
3. Use the specific sample that will be provided for your experiments.
4. Use your code to create a new GeoJSON that includes a polygon delineating your home country. Visualize it (e.g. by opening it via the github interface).

Sneak peak at your exercise

Example

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pythex

A Python `re` module tester that runs entirely in your browser.

Run: 41.0 ms

Copy link to this regex

Your regular expression:

```
(?P<the_one>line one)|(^line two$)|(line three(.+)line four)
```

Your test string:

```
line one
line two
line three
line four
LINE ONE
```

Show Python source

Flags

IGNORECASE MULTILINE DOTALL VERBOSE ASCII DEBUG

Mode

finder (all) search (first) fullmatch

Match result:

```
line one
line two
line three
line four
LINE ONE
```

Match captures:

Match 1

```
"line one"
```

Numbered Groups:

- Group 1: line one
- Group 2: Empty
- Group 3: Empty
- Group 4: Empty

Named Groups:

- the_one: line one

Regular Expression Cheatsheet

Inspired by Rubular. For a complete reference, see the official [re module documentation](#).

Made by Gabriel Rodríguez. Running Python 3.13.2 in the browser via Pyodide.

```
{ "type": "FeatureCollection",
  "features": [
    { "type": "Feature",
      "geometry": { "type": "Point", "coordinates": [102.0, 0.5] },
      "properties": { "prop0": "value0" }
    },
    { "type": "Feature",
      "geometry": {
        "type": "LineString",
        "coordinates": [
          [102.0, 0.0], [103.0, 1.0], [104.0, 0.0], [105.0, 1.0]
        ]
      },
      "properties": {
        "prop0": "value0",
        "prop1": 0.0
      }
    },
    { "type": "Feature",
      "geometry": {
        "type": "Polygon",
        "coordinates": [
          [ [100.0, 0.0], [101.0, 0.0], [101.0, 1.0],
            [100.0, 1.0], [100.0, 0.0] ]
        ]
      },
      "properties": {
        "prop0": "value0",
        "prop1": { "this": "that" }
      }
    }
  ]
}
```

Sneak peak at your exercise

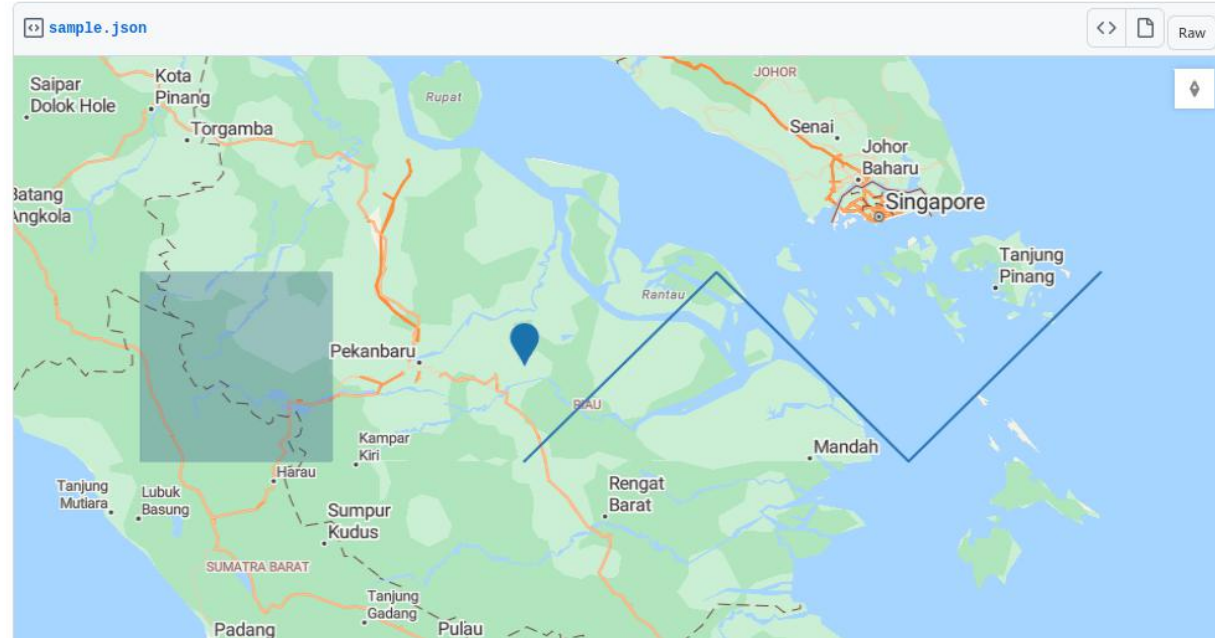
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Sample GeoJSON File



Thank you!

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